

Md Fahim Sarker Mridul

AI/ML Engineer · Python Developer

fahimsarker0805@gmail.com · +880 1400-530058 · Portfolio · GitHub · LinkedIn

Education

Green University of Bangladesh

Feb 2022 – Jan 2026

Bachelor of Science in Computer Science & Engineering

Major: Artificial Intelligence and Machine Learning

Experience

SparkTech Agency

Sept 2025 – Present

Mid-Level AI Engineer

July 2026 – Present

Junior AI Engineer

Sept 2025 – June 2026

- **Technical Leadership:** Promoted to direct a cross-functional team of 3–4 engineers, overseeing end-to-end system design, client communication, and the successful delivery of 10+ production AI systems (RAG, OCR, NLP, CV).
- **Scalable AI Architecture:** Engineered high-performance RAG pipelines using LangChain, FAISS, Pinecone, and FastAPI, optimizing vector retrieval and LLM orchestration to achieve sub-2-second latency.
- **MLOps & Infrastructure:** Established automated MLOps workflows via Weights & Biases and AWS, accelerating model experimentation cycles and streamlining CI/CD for production deployments.

GUB Mars Rover Team

Aug 2024 – Oct 2025

Software Engineer

- Developed and programmed the hardware systems of CLADOS, GUB Mars Rover Team's primary rover, which ranked 6th worldwide and 1st in Asia at the 2026 *Anatolian Rover Challenge*.

Projects

Right Route App

getrightroute.app

AWS, FastAPI, OpenAI, NLP

- Designed the system architecture and engineered an AI-powered Route-Processing pipeline for the USA using AWS Textract, OpenAI, NLP, and FastAPI for automated route extraction from permit documents.

Legal RAG Assistant

casezys.com

LangChain, FAISS, FastAPI, OpenAI

- Developed a production-grade legal assistant for U.S. Federal and Michigan State law using semantic retrieval and LLM-based reasoning. Designed article-level indexing and retrieval pipelines supporting accurate legal document search and question answering.

Daily Storybook AI

Case Study

FastAPI, Microservices, Event-Driven, AWS, PostgreSQL, EC2

- Architected and developed a scalable, event-driven backend for an AI-powered Fitness Storybook platform using Python (FastAPI), PostgreSQL, and AWS (ECS, SQS, S3, Bedrock), decoupling heavy AI workloads from the main API to eliminate timeouts and improve system reliability.

Open Source Contribution

NeuroGebra

PYPI · Documentation · DOI

Real Training with full ML Transparency

- Developed an open-source symbolic-neural framework for visualizing machine learning mathematics and neural network operations in real time. Adopted by **8000+ students and researchers worldwide** as a learning companion alongside PyTorch and TensorFlow.

Technical Skills

Programming Languages: Python, Java, C, SQL

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Feature Engineering

Gen AI & NLP: LLMs, LangChain, RAG, HuggingFace Transformers, PEFT, LoRA, Sentiment Analysis

Computer Vision: OpenCV, YOLO, Vision Transformers, CNNs, Image Segmentation, Object Detection, XAI

MLOps & Deployment: FastAPI, REST APIs, Docker, CI/CD, MLflow, Weights & Biases (W&B)

Cloud Platforms: AWS EC2, AWS S3, AWS Textract, AWS Bedrock, AWS SageMaker, Google Cloud Services

Data Science & Analytics: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Statistical Analysis, Data Visualization

Tools & Infrastructure: Git, GitHub, Linux, Jupyter Notebook, Google Colab, Kaggle, Postman

Involvement & Awards

Comilla Univ. Data Science Hackathon: 1st Runner-Up (Nov 2024) out of numerous participants.

IDP-1 University Project Showcase: Awarded Best Innovation Award for ViolaShield (Jan 2025).

GUB Computer Club: Information Secretary (2023–2024), organized 5+ technical workshops on ML for 200+ students.

BASIS Students' Forum: Enduring Member, mentoring current members on industry practices and career development.

NSU FutureBuilders AI Hackathon: Finalist (Dec 2024) | BUBT AI Competition Finalist (Nov 2025).